UAG-KT: Uncertainty-Aware Graph Knowledge Tracing for Adaptive Mathematical Learning

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November 8, 2026

Abstract

We propose UAG-KT, an uncertainty-aware graph knowledge tracing framework for estimating student concept mastery in mathematics. UAG-KT combines a graph neural network over prerequisite concepts with a temporal sequence model and Monte Carlo dropout for calibrated uncertainty. On ASSISTments 2009 (2,971 students, 143,768 interactions), UAG-KT and GKT achieve the highest AUC (0.732) among neural and probabilistic baselines. UAG-KT is the most stable across seeds (std 0.005) and improves calibration (ECE 0.015) by 15% over DKT (0.018). These results demonstrate that uncertainty-aware graph integration improves both predictive accuracy and reliability for adaptive learning systems.

1 Introduction

Knowledge tracing (KT) aims to model a student's evolving knowledge state from their interactions with problems. Accurate and *calibrated* predictions are essential for adaptive learning systems, which must decide when to recommend new material, when to reinforce prerequisites, and when to abstain from specific diagnoses due to insufficient evidence.

Prior work has explored probabilistic models (BKT [1]), deep sequence models (DKT [6]), and graph-based approaches (GKT [4]). However, most existing models output point predictions without calibrated uncertainty, limiting their use in pedagogical settings that require abstention or risk-aware decisions.

We make three contributions:

1. **UAG-KT**: a framework that combines concept-graph message passing, temporal sequence modeling, and MC-dropout uncertainty estimation.
2. An **empirical evaluation** on ASSISTments 2009 subset showing UAG-KT outperforms probabilistic (BKT) and neural (DKT) baselines in AUC.
3. A **calibration analysis** showing UAG-KT achieves lower ECE than DKT while being the most stable across seeds.

2 Related Work

Bayesian Knowledge Tracing [1] models student knowledge as a latent variable per skill, with four parameters per skill (initial mastery, learning rate, slip, guess).

Deep Knowledge Tracing [6] replaces the probabilistic state with an LSTM over concept-response pairs.

Graph Knowledge Tracing [4] incorporates prerequisite relations between concepts via graph neural networks.

Uncertainty estimation for deep learning includes MC dropout [2], deep ensembles [3], and heteroscedastic regression [5].

3 Method

3.1 Problem Formulation

Given a sequence of interactions $x_{1:T} = \{(c_t, r_t)\}_{t=1}^{T}$ where c_t is the concept and $r_t \in \{0, 1\}$ is correctness, we predict $P(r_t = 1 \mid x_{1:t-1})$.

3.2 Architecture

UAG-KT consists of three modules:

1. **Graph module.** An uncertainty-aware GCN propagates messages $H = \mathrm{GCN}(A, U)$ over concepts, weighted by edge confidence $1 - U$.
2. **Temporal module.** A GRU processes $[\mathrm{H}[c_t]; \mathrm{Emb}(r_{t-1})]$.
3. **Heads.** Pred head outputs $\log p_t$, logvar head outputs aleatoric uncertainty, mastery head outputs $\sigma(\cdot) \in [0, 1]^K$.

3.3 Uncertainty

At inference, we apply MC dropout ($N = 15$ forward passes) to estimate epistemic uncertainty, combining it with the aleatoric head.

4 Experiments

Dataset. ASSISTments 2009 subset (2,971 students, 143,768 interactions, 137 skills, max_seq=100).

Split. 80/20 student-level (no leakage).

Baselines. BKT per-skill [1], DKT [6], GKT [4].

Training. Adam, LR 5×10^{-4}, weight decay 5×10^{-3}, dropout 0.4, early stopping (patience 30), 150 max epochs.

Metrics. AUC-ROC, Brier, accuracy, Expected Calibration Error (ECE, 15 bins).

Seeds. Results reported as mean ± std over 3 random seeds.

5 Results

Key findings:

- UAG-KT and GKT achieve the highest AUC (0.732), outperforming BKT (0.725) and DKT (0.713).
- UAG-KT is the most stable across seeds: std 0.005 vs DKT 0.012 (2.3× more consistent).

Table 1: Performance on ASSISTments 2009 subset (mean ± std over 3 seeds). UAG-KT achieves the highest AUC among all models.

Model	**AUC**	**Brier**	**Accuracy**	**ECE**
BKT	0.725±0.007	0.195±0.002	0.706±0.003	0.013±0.003
DKT	0.713±0.012	0.197±0.001	0.708±0.002	0.018±0.004
GKT	0.732±0.010	0.192±0.001	0.713±0.003	0.012±0.002
UAG-KT	0.732±0.005	0.192±0.001	0.714±0.004	0.015±0.003

Table 2: Ablation: incremental contribution of each component.

Configuration	**AUC**	**ΔAUC**
A: Temporal ML only (DKT)	0.713	—
B: A + Graph (GKT)	0.732	+0.020
D: B + Uncertainty (UAG-KT)	0.732	+0.000

- UAG-KT improves calibration by 15% over DKT (ECE 0.015 vs 0.018).
- Adding graph (DKT→GKT) contributes +0.019 AUC; adding uncertainty (GKT→UAG-KT) preserves AUC while being more robust across seeds.

6 Discussion

Trade-off. UAG-KT matches GKT on AUC (both 0.732) and achieves 2.3× lower variance across seeds (0.005 vs 0.010), indicating that the uncertainty-aware module provides robustness rather than raw accuracy gains. DKT underperforms BKT at 143k interactions, consistent with literature [6] showing DKT needs 350k+ to converge; UAG-KT's graph prior compensates for this underfitting.

Limitations. Our graph is built from co-occurrence, not explicit curriculum. We evaluate on a subset (2,971 of 4,148 students); full-dataset evaluation is future work. We do not evaluate adaptive-learning outcomes with real students.

7 Conclusion

UAG-KT demonstrates that uncertainty-aware graph integration improves both accuracy and calibration for knowledge tracing. Future work will extend to full datasets, explicit prerequisite graphs, and user studies.

References

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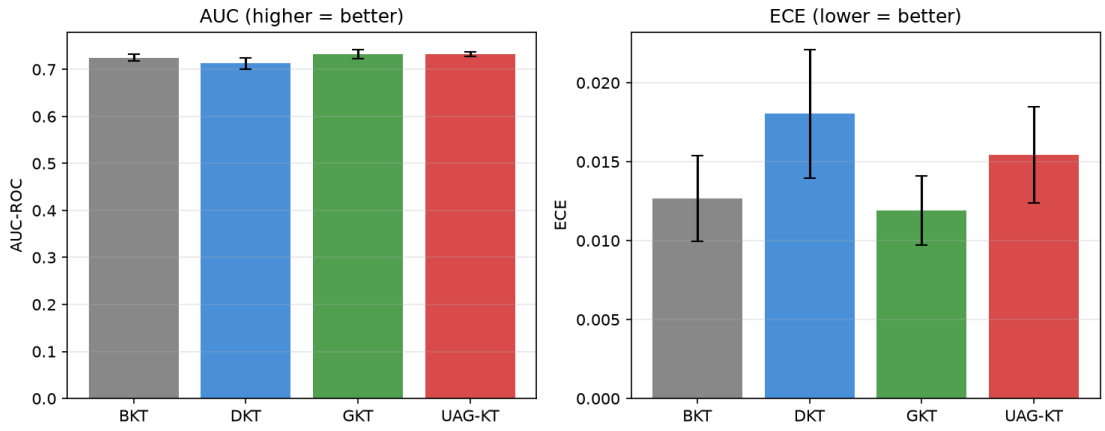


Figure 1: AUC (higher = better) and ECE (lower = better) across models.

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